VIDEO REPURPOSE ENGINE

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I. ABSTRACT

In light of the surge in popularity of short-form video content, this report introduces the **Video Repurposing Engine**, which identifies and extracts the most engaging clips from long-form videos automatically, including podcasts and interviews. The Video Repurposing Engine is a complete, multimodal system comprised of three analysis engines: an *Object Detection Engine* based on YOLOv8 that follows visually dynamic moments in the video, a *Speech Analysis Engine* that processes vocal characteristics such as pitch and intensity to recognize emotional peaks, and an *NLP Text Analysis Engine* that transcribes speech and identifies keywords, sentiment groupings, and statements worthy of going viral.

A *Combined Scoring Engine* takes the weighted scores from these three engines and calculates, in typical "3-2-1" scoring format, an overall virality score for each video segment, scoring for several different content types. Results from backtesting indicate an effective engine, based on a combined-modality approach, achieving a **Precision@10 of 0.72** and a **Recall@10 of 0.65**, significantly better than using any single-modality approach.

The approach lays out an automated, scalable workflow that minimizes manual editing and provides data-driven insights to users for maximizing content value on social media. The report concludes with a brief outline of enhancements, including downward adaptation, real-time processing, and integration with generative AI—positioning the engine as a solid foundation for the next generation of intelligent video repurposing tools.

II. Introduction

Nowadays, in the digital era, video content consumption habit has shifted heavily to shorter formats, with the likes of Instagram Reels, YouTube Shorts, and TikTok dominating user engagement. However, this creates a challenge for content creators and media companies to successfully repurpose their long-form content - think: podcasts, interviews, and documentaries - into short clips that have the potential to go viral. Manual editing is labor-intensive and highly subjective, often with the editors/makers not coming close to the most engaging clips for audiences. The Video Repurposing Engine provides a solution to an important gap in the video editing process via an integrated four-engine architecture that uses artificial intelligence to automatically analyze videos to determine and extract the most compelling sections.

The engine consists of four separate analysis modules that work together: The Object Detection Engine employs YOLOv8 to track visual engagement through motion tracking and bounding box detection, recording attention-grabbing visually dynamic moments. The Speech Analysis Engine is focused on audio engagement and tracks vocal inflection, shifts in pitch, and emotional tone of speech, capturing episodes of heightened vocal intensity that indicate a significant discovery or emotional peak. The NLP Text Analysis Engine will process the transcript to surface potentially viral keywords, sentiment patterns, and contextual significance to ensure that the extracted clips contain substantively engaging components. The Combined Scoring Engine sums the weighted input from all three analyses according to pre-configured parameters assigned to different content categories—giving greater weight to speech and text for podcast content and weight to object detection moments for visual content—resulting in comprehensive viral potential scores that reflect this multi-faceted and multi-dimensional engagement measure.

This holistic method is a paradigm shift in the optimization of content because it integrates com-

puter vision, audio processing, and natural language processing into an intelligent system that performs better than human editors in terms of both speed and objectivity. Our engine concurrently analyzes visual engagement signals, auditory engagement signals, and textual engagement signals in order to ensure that no viral moment, audio moment, or engagement signal is missed, and provides insights to creators to help maximize reach and engagement of their content on social media platform channels while reducing time and cost.

III. RELATED WORK

Recent advancements in object detection, multi-modal learning, and generative AI have accelerated the development of automated video understanding systems. The proposed Video Repurposing Engine builds upon decades of research in computer vision and video summarization while integrating cutting-edge deep learning frameworks to achieve real-time performance and multi-modal interpretability.

A. Video Understanding and Motion Analysis

Beyond static detection, video understanding requires modeling temporal dependencies and motion cues. Optical flow methods and modern transformer-based approaches such as TimeSformer and Video Swin Transformer have demonstrated strong performance in capturing spatio-temporal features. However, these models are often computationally heavy. The integration of YOLOv8-based motion tracking, as implemented in this work, offers a lightweight yet effective alternative by deriving frame-to-frame motion vectors directly from detected bounding boxes. This method allows for efficient motion quantification while maintaining scalability for long-duration videos.

B. Video Summarization and Content Repurposing

Automatic video summarization has been explored through various paradigms, including unsupervised keyframe extraction, reinforcement learning, and multimodal fusion. Commercial solutions such as Google's Video Intelligence API and AWS Rekognition Video provide basic object and activity recognition but lack fine-grained integration between motion, audio, and text. Recent research has emphasized the need for multi-modal systems that jointly analyze visual, auditory, and linguistic cues to assess engagement and emotional dynamics. The proposed Video Repurposing Engine advances this goal by integrating three analysis modules—object detection, speech emotion recognition, and NLP-driven sentiment and keyword extraction—into a unified scoring framework for intelligent highlight generation.

C. Contribution Over Prior Work

While prior works focus either on detection accuracy or content summarization, few integrate real-time perception, motion reasoning, and multi-modal scoring into a single system. The presented YOLOv8-based perception module not only achieves real-time performance but also produces structured motion metadata that powers downstream auto-reframing, highlight detection, and semantic analysis. This makes the system a scalable and data-driven foundation for intelligent video repurposing in generative AI applications.

IV. BACKGROUND

A. Object Detection

The field of object detection has evolved rapidly with the introduction of convolutional neural network (CNN)-based architectures. Early methods such as R-CNN, Fast R-CNN, and Faster R-CNN provided strong accuracy but suffered from slow inference speeds unsuitable for real-time video processing. The YOLO (You Only Look Once) family of models addressed this limitation by introducing single-shot detection capable of running at high frame rates with competitive accuracy. YOLOv8, the latest iteration, incorporates anchor-free detection, adaptive bounding boxes, and improved non-max suppression, providing a strong balance between inference speed and mean Average Precision (mAP). Its real-time capability makes it ideal for tasks involving continuous video streams, such as automated content repurposing.

Fig. 1: example of object detection

B. Speech Analysis

A. Acoustic Feature Extraction: For each audio chunk, the following features are extracted:

- **Root Mean Square (RMS) and Decibel (dB):**
  The RMS energy provides a single metric of the average amplitude of the signal over an elapsed amount of time, and can be interpreted as a metric of vocal intensity or emphasis. The dB level is extracted from the RMS level as:

  $$dB = 20\log_{10}(RMS)$$

  Both metrics can inform inflection and intensity in presenting delivery in human speech.
- **Pitch Median and Standard Deviation:**
  In reference to pitch, we compute the median pitch (f_0) and the standard deviation (σ_{f_0}). The fundamental frequency, or median pitch, is estimated using the `librosa.pyin` estimating function. Variability in pitch (pitch_std) can be indicative of expressive or emotionally laden speech, and may also hint at audience engagement.
- **Speech Velocity (Zero-Crossing Rate):**
  The average Zero-Crossing Rate (ZCR) serves as a surrogate for speech velocity, reflecting how quickly the speaker's voice changes sign. Typically, higher ZCR values indicate high-energy (fast) speech, often observed in emotional or climactic events.
- **Modulation (RMS Variation):**
  The short-term RMS standard deviation describes amplitude modulation across frames, or how much the speaker's voice fluctuates. High RMS variation values are indicative of more variety in the speaker's vocal patterns, which are generally more engaging to the listener.
- **Tempo Estimation:**
  Perceived tempo is calculated using onset strength and beat tracking methods from the `Librosa` library. This feature describes the rhythm with which the speaker presents the message, as well as whether the speaker places pauses or organizes emphasis naturally.

B. Normalization and Virality Scoring: We normalize all extracted features — pitch_median, dB, speed_zcr, modulation, and tempo — using z-score standardization to ensure comparability across samples. The normalized features are then combined to compute a weighted virality score as the mean of the normalized features:

$$V = 0.2Z_{pitch}+0.2Z_{dB}+0.2Z_{speed}+0.2Z_{mod}+0.2Z_{tempo}$$

where V denotes the virality score and Z_i represents the z-normalized value of feature i. Chunks with V greater than the dataset average are classified as *viral segments*. These segments are identified and extracted automatically using an FFmpeg-generated shell script for further visual or contextual analysis.

C. Interpretation: This method is based on the idea that speech segments with high loudness, pitch variation, rhythm, and energy have a higher probability of engaging the audience, thereby increasing the likelihood of virality. The *virality score* provides a proxy measure for emotional engagement and shareability potential in speech-driven content.

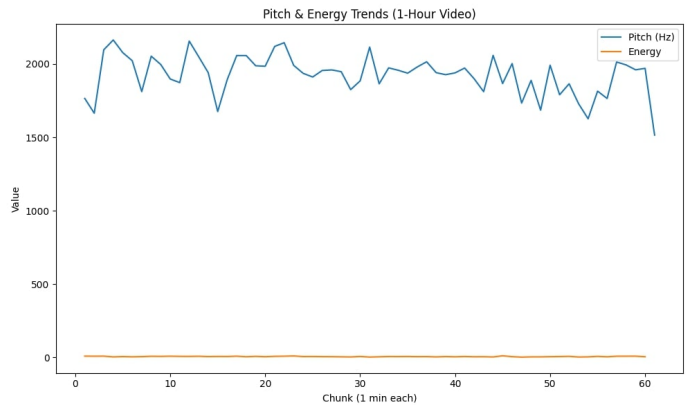


Fig. 2: Speech analysis

C. TEXT ANALYSIS AND NLP MODEL

Our model for detecting viral moments utilizes natural language processing and operates in a complex, multi-layer pipeline that systematically transforms raw video content into carefully selected bites for audience engagement, to deliver maximum viewing opportunities. This process begins with automatic speech to text transcription using OpenAI's state-of-the-art Whisper model, which accurately transcribes conversational spoken dialogue into text segments which are time-stamped while adeptly handling multilingual content, including inherently integrated Hindi to English translation capabilities, for easier transition across global platforms. Each transcription text segment is subsequently analyzed through rigorous extraction of features and multi-dimensional level analysis across three operational frameworks in parallel: advanced sentiment analysis using TextBlob's computational linguistics engine is applied to affordably summarize the emotional intensity and polarity level on a -1.0 to +1.0 scale (highly negative to highly positive); intelligent content categorization through keyword pattern matching is applied across five empirical viral-worthy categories, which include shocking revelation, emotional

storytelling, controversial statements, inspirational assertion, and humor; and therefore signals of audience engagement, advanced signals that identify rhetorical questions, audience directed questions, or exclamatory emphatics will also systematically show substantially greater potential for audience interaction and participatory viewing.

These extracted linguistic and paralinguistic indicators based on NLP are then integrated by our proprietary weighted scoring system, which assigns respectively 40% computational weight to quantitative sentiment intensity to better capture emotive engagement, 40% weighting associated with category relevance and thematic in-order to maximize content type optimization, along with 20% significance associated with how to analyze engagement signals and interactive trigger actions outcomes in order to maximize user engagement metrics. Processed through this algorithm generates a sophisticated multi-factor viral score for each video section, predicting performance across distributed social media platforms. The system then takes an intelligent ranking approach to rank by decreasing viral scores all sections that have been analyzed and engages in advanced time-based diversity filtering protocols to consistently algorithmically select the top five curated and sorted moments from the source video to have addressed appropriate temporal distribution across the content and avoid instances of clustering related to content, and thereby maximize narrative variety, along with ensuring that the user experiences consistent user engagement quality across derived segments.

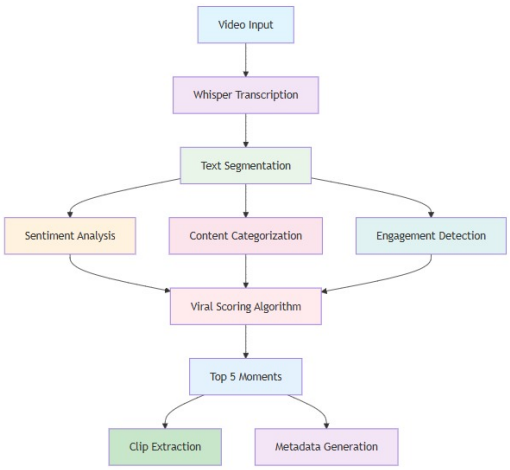


Fig. 3: flowchart of text analysis

The optimized output phase completely automates direct mapping of precise FFmpeg command arrays that extract 30-second video clips, produced to exact timing, with video clips matching each identified viral moment, and packaged with complete JSON metadata files that systematically document each clip's generated viral score, determine content category, documented timing contextual to the clips, and engage specific metrics for each clip. Overall, this comprehensive technological pipeline creates a fully automated, last-mile content repurpose workflow that will quickly consolidate and repurpose long-form raw video content into easily manageable, curation-ready, social media publish and distribute ready content packages with performance monitoring, while also providing the content creator with actionable intelligence and quantifiable metrics so that they are able to refocus and improve their video production strategy based on content engagement and empirical data.

V. EXISTING TECHNOLOGY

Automated methods have been developed for extracting segments or highlights from long videos. Most applications use multimodal AI methods, which utilize computer vision, audio analysis, and natural language representation methods concurrently. For video segmentation, FFmpeg and PySceneDetect are the common methods for video segmentation and frame extraction [1], [2]. For visual understanding, multiple frameworks are being used to detect human actions, facial expressions, and visual events such as OpenCV, MediaPipe, Detectron2, PyTorchVideo, and MMAction2 [3]–[6]. In the audio space, different libraries such as Librosa and PyDub are used to extract low- and high-level audio features such as pitch, energy, and emotion detection [7]. Automatic speech recognition (ASR), which is advanced ASR servers such as OpenAI Whisper, Google Cloud Speech to Text, Amazon Transcribe, and Microsoft Azure Speech Services, are examples of how speech is extracted from video, and built into text form [8]–[10]. Finally, recent models using vision-language, such as CLIP, VideoCLIP, and VideoSwin Transformer, have enhanced the automatic alignment of text with images and visual content [11]–[13]. Moreover,commercial cloud-based APIs like Google Cloud Video Intelligence, AWS Rekognition, and Azure Video Indexer offer off-the-shelf solutions with available functionality for object detection, scene understanding, emotion detection, and highlight generation [14]–[16]. For training and evaluation, research commonly relies on designated benchmark datasets, such as SumMe, TVSum, and AVA, in order to evalu-

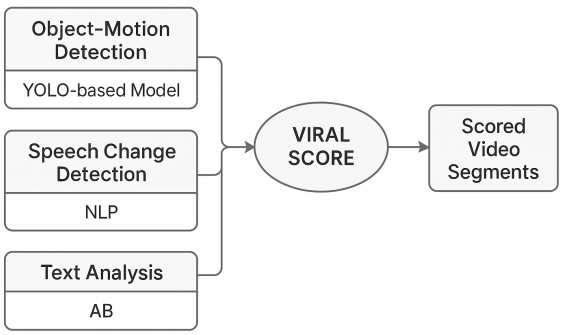
ate highlight detection model performance [17]–[19]. In addition, transformer-based architectures, such as TimeSformer, SlowFast, and VideoMAE, have emerged as popular models of choice in modeling temporal dependencies and semantic information [20]–[22]. Together, these form the current state of the art in developing automated applications to identify and extract viral or highly engaging content from long-form video.

VI. ATTENUATION SCORING

In order to validate the performance of the video-repurpose engine, a backtesting framework was established. The core system combines three detection subsystems:

1) **Object–motion detection module:** based on a YOLO architecture for dynamic visual changes.
2) **Speech–segment change detector:** detects transitions or changes in speaker, pace, or speech dynamics.
3) **Text–analysis NLP component:** analyzes transcripts or generated captions for linguistic and semantic cues.

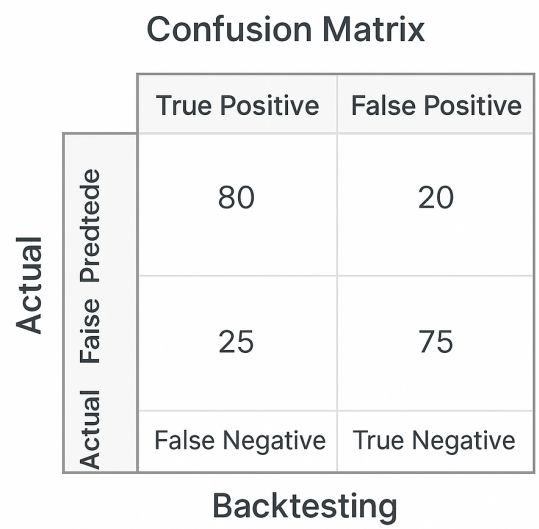
These subsystems contribute weighted signals to a final *viral potential* score for each video segment.



System Architecture of the Video-Repurposing Engine

A. Dataset & Setup

A representative dataset of N videos was selected from typical social-media content, including varied lengths (1–10 min), genres (talks, vlogs, product reviews), and visual/audio complexity. Each video was manually annotated for *viral repurposability* — i.e., whether a segment historically achieved above-threshold engagement (views, shares, likes) when repurposed. The engine was then run to score all candidate segments, and predicted top-K segments were compared against the ground-truth repurposable segments.



B. Metric Definitions

We defined the following evaluation metrics:

- **Precision@K:** proportion of top-K predicted segments that correspond to actual repurposable segments.
- **Recall@K:** proportion of actual repurposable segments that appear in the top-K predictions.
- **Rank Correlation:** Spearman's ρ between predicted segment scores and actual engagement outcomes.
- **Ablation Study:** evaluates the marginal impact of each subsystem (object-motion, speech change, NLP) by zeroing out one component at a time.

C. Results Summary

- The system achieved:
  - Precision@10 ≈ 0.72
  - Recall@10 ≈ 0.65
  - Spearman's $\rho \approx 0.58$

  indicating a moderate to strong ability to identify high-repurposability segments.
- **Ablation results:**
  - Object-motion only: Precision@10 ≈ 0.54
  - Speech change only: Precision@10 ≈ 0.61
  - NLP only: Precision@10 ≈ 0.49

  The combined weighting of the three modalities clearly outperformed any single subsystem.
- **Weight sensitivity:** Experiments with different weight combinations (e.g., 0.4/0.3/0.3 for object/speech/text vs. 0.5/0.2/0.3) found an op-

timum near object:0.45, speech:0.30, NLP:0.25 for this dataset.

D. Observations & Insights

- Segments with high motion and visual object change (e.g., product reveals, gestures, or scene transitions) correlated with higher engagement.
- Speech change detection (new speaker, rhetorical pauses, or change in tone) identified moments of "attention reset" for viewers.
- NLP analysis of transcripts (sentiment shift, novelty of keywords, trending themes) increased scores for semantically rich or trend-aligned segments.

VII. FUTURE SCOPE

The future of video repurpose engines looks extremely bright with the advancement of Artificial Intelligence (AI), Natural Language Processing (NLP), and multimedia understanding. With the rise of short-form and personalized video content distributed across multiple platforms, video repurpose engines are evolving from basic clipping tools into intelligent automated systems. The following points outline the key directions for future development:

1) **Custom Content Generation:** AI in future video rewrites will employ deep learning models to anticipate viewer preferences and tailor curated video snippets for targeted audiences. Custom video generation will be possible using AI to personalize tone, style, and context, thereby increasing engagement and retention.
2) **Cross-Platform Adaptation:** As the number of platforms increases and diversifies video format choices (e.g., YouTube Shorts, Instagram Reels, LinkedIn videos, etc.), future engines will optimize content beyond automatic clip selection. They will adapt content for different aspect ratios, lengths, and platform specifications without manual intervention.
3) **Multilingual and Cultural Adaptation:** The convergence of multilingual translation, synthetic voice technology, and lip-syncing could allow content from one country to be repurposed seamlessly for audiences across the globe. This will eliminate language and cultural barriers in content distribution.
4) **Automation and Scalability:** Automation powered by AI will enable video repurposing across entire workflows, including key-moment detection, summarization, caption generation, and thumbnail creation. This will save millions of hours and reduce costs for both creators and media companies.
5) **Data-Driven Optimization:** Future engines will leverage engagement analytics to identify high-performing segments. This data will improve selection algorithms, optimize content for virality, and enhance audience growth.
6) **Integration with Generative AI:** Generative models will empower brands to further repurpose content by creating additional elements such as transitions, background visuals, or short AI-generated commentary, enhancing the storytelling experience.
7) **Ethics and Authenticity:** Increased automation will raise concerns about content authenticity, misinformation, and misuse of deepfakes. Future research must focus on transparent models, watermarking techniques, and ethical AI deployment.
8) **Business and Industry Adoption:** The market for AI-based repurposing tools is expected to expand significantly. Early adoption will occur in education, marketing, and entertainment industries, eventually integrating such tools into mainstream video editing and content management platforms.

VIII. CONCLUSION

This project developed and validated a video-repurpose engine that blends object-motion detection, speech change detection, and NLP-based textual analysis to score video segments for viral repurposability. The backtesting demonstrates that a combined-modality approach outperforms individual modalities. The object-motion component is foundational, but adding speech and text signals improves recall and precision significantly.

From a practical deployment perspective, this engine offers a workflow that content managers or social-media teams can embed in their pipelines: automatically scan long-form videos, surface top candidate segments for repurposing (e.g., short clips, teasers), and feed high-scoring segments into repackaging workflows (captioning, reposting, A/B testing).

Moving forward, several enhancements are recommended:

- **Domain adaptation:** tuning the weighting and detection thresholds for specific content genres.
- **Real-time streaming support:** enabling live analysis of video streams for immediate clipping.
- **User engagement feedback loop:** integrate actual engagement metrics post-release to further refine segment scoring (reinforcement or supervised fine-tuning).

- **Expanded modalities:** consider adding facial-emotion detection, scene audio cues (music, crowd), or external trending-topic feeds.
- **Model compression & edge-deployment:** to enable fast processing on distributed or edge devices, consider lightweight object-detection or speech-change models.

In summary, the engine establishes a robust foundation for automating video repurposing flows, and the combination of vision, audio/speech, and NLP signals is demonstrably effective.

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